



Mark Scheme (Results)

January 2022

Pearson BTEC Nationals

In Applied Science (31617H1P)

Unit 1: Principles and Applications of Science I -
Physics

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Unit 1: Principles and Applications of Science I

General marking guidance

- All learners must receive the same treatment. Examiners must mark the first learner in exactly the same way as they mark the last.
- Marking grids should be applied positively. Learners must be rewarded for what they have shown they can do, rather than be penalised for omissions.
- Examiners should mark according to the marking grid, not according to their perception of where the grade boundaries may lie.
- All marks on the marking grid should be used appropriately.
- All the marks on the marking grid are designed to be awarded. Examiners should always award full marks if deserved. Examiners should also be prepared to award zero marks, if the learner's response is not rewardable according to the marking grid.
- Where judgement is required, a marking grid will provide the principles by which marks will be awarded.
- When examiners are in doubt regarding the application of the marking grid to a learner's response, a senior examiner should be consulted.

Specific marking guidance

The marking grids have been designed to assess learner work holistically. Rows in the grids identify the assessment focus/outcome being targeted. When using a marking grid, the 'best fit' approach should be used.

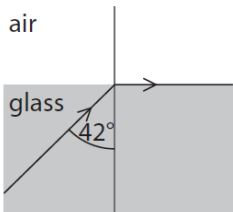
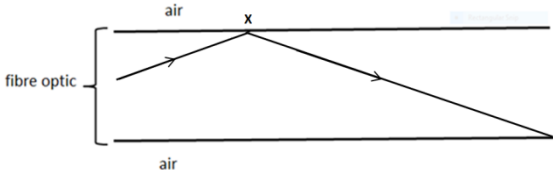
- Examiners should first make a holistic judgement on which band most closely matches the learner's response and place it within that band. Learners will be placed in the band that best describes their answer.
- The mark awarded within the band will be decided based on the quality of the answer, in response to the assessment focus/outcome and will be modified according to how securely all bullet points are displayed at that band.
- Marks will be awarded towards the top or bottom of that band, depending on how they have evidenced each of the descriptor bullet points.

BTEC Next Generation Mark Scheme

Question Number	Answer	Additional Guidance	Mark
1 (a)(i)	19 (cm)	accept 18 (cm) to 20 (cm) inclusive	1
1 (a)(ii)	80 (cm)	78 (cm) to 82 (cm) inclusive	1
1 (b)	substitution (1) $(v=) 16.0 \times 1.8$ evaluation (1) $29.0 \text{ (m s}^{-1}\text{)}$	evaluation 29 (m s ⁻¹) (2 marks) correct answer with no working gains full marks accept 28.8 (m s ⁻¹) accept 28.8 or 29.0 to any other power of 10 for 1 mark allow ecf for 1 mark	2
1 (c)	B 90°		1
1 (d)	First sentence perpendicular/at right angles / 90° (1) Second sentence parallel/ in the same direction (1)	accept 90 degrees /normal to /up and down accept to and fro/ backwards and forwards accept side to side once	2
Total			7 marks

Question Number	Answer	Additional Guidance	Mark
2 (a)(i)	C 101		1
2 (a)(ii)	Award one mark for identification and one mark for an appropriate expansion from the same pair PAIR 1 Identification (1) (digital signals) can travel further /long distance Expansion (1) (because) (they)can be regenerated (1) PAIR 2 Identification (1) Quality (of signal) is maintained/better/clearer Expansion (1) (because there is) {less noise/less energy loss/less attenuation/less interference/less distortion/less information loss} PAIR 3 Identification (1) signals can be encrypted Expansion (1) gives greater security/more difficult to intercept/interfere with/hack PAIR 4 Identification (1) can be filtered/restored/cleaned up /prevent data loss Expansion (1) (because) binary code/1s and 0s (can be easily identified) PAIR 5 Identification (1) more information/data can be transferred/transmitted/travel each second Expansion (because) bandwidth is greater/ multiplexing	Ignore quicker/faster accept more information /data can be transferred faster/quicker	2

2 (b)	A description including the following: (computer picks up a signal) from {router/mobile phone/dongle/mobile hotspot /hub/Wi-Fi box/modem} (1) {router/mobile phone/dongle/mobile hotspot /hub/Wi-Fi box/modem phone} connected to broadband/ internet (1) (connections given by) telephone cables/optical fibres/satellite links/radio transmission/ handshake (1)	reverse order is acceptable	3
Total			6 marks

Question Number	Answer	Additional Guidance	Mark
3 (a)	C 		1
3 (b)	 Continuous reflected ray from X to the right of X within the optical fibre (1) TIR that shows angle of incidence equals angle of reflection (1)	Diagram completed as shown gains 2 marks accept line without arrow do not accept dotted line independent mark correct angle for TIR takes reflected ray to a point at the end of the bottom edge of the optical fibre, as shown in the diagram	2

		allow 1 cm to right or left of this point, if line does not reach bottom edge judge by eye	
3 (c)	endoscopy/colonoscopy/bronchoscopy /laparoscopy/keyhole surgery/see inside the (human) body Allow any other correct medical application.	accept (see inside) any named part of the body ignore scan(s)	1
Total			4 marks

Question Number	Answer	Additional Guidance	Mark
4 (a)	Award one mark for identification and one additional mark for each appropriate expansion up to a maximum of three marks. <i>Identification (any one from)</i> wavelength is longer/double (in a closed pipe/ in 7b) (1) <i>expansion (any two from)</i> wavelength 2L in open pipe (1) wavelength 4L in closed pipe (1) wavelength shows first harmonic in closed pipe (1) velocity of sound is the same in both pipes (1) use of $f = v/\lambda$ / $f=v/2L$ for open pipe / $f=v/4L$ for closed pipe (1)	accept reverse argument throughout accept greater/bigger/higher	3
4 (b)	substitution (1) $\frac{1 \times (10^{-5})}{5 \times (10^{-10})} = \left(\frac{D_2^2}{2.5^2} \right)$	answer 350 (m) for 4 marks correct answer with no working gains full marks substitution and rearrangement in either order	4

	rearrangement (1) $(D_2)^2 = \frac{1 \times (10^{-5}) \times 2.5^{(2)}}{5 \times 10^{-10}}$ evaluation (1) 125000 evaluation of square root (1) 350 (m) Method using constant, k $I = \frac{k}{D^2}$ substitution (1) $1 \times (10^{-5}) = \frac{k}{6.25}$ evaluation of k (1) $k = 6.25 \times (10^{-5})$ substitution and rearrangement (1) $5 \times (10^{-10}) = \frac{6.25 \times (10^{-5})}{D^2}$ $D^2 = \frac{6.25 \times (10^{-5})}{5 \times (10^{-10})}$	20000 or 6.25 seen gains a mark 50000 from use of 2.5 in rearrangement gains 2 marks any additional incorrect working after 125000 ignore gains 3 marks accept any value that rounds to 350(m) 224(m) gains 3 marks use of 25 not 25² 559(m) gains 3 marks intensities inverted	
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	evaluation and square root (1) $D^2 = 1.25 \times 10^5$ $D = 350(m)$		
Total			7 marks

Question number	Indicative content
5	Advantages • higher bandwidth/frequency/ shorter wavelength • allows large amounts of data to be transmitted • microwaves can be split into channels • multiplexing (split into many frequencies) • very little interference (between signals) • rapid connection (to internet using mobile networks) • signal can be directed to specified location • antenna size is small • can connect to other devices • must be within range of a base station • can pass through walls Disadvantages • quality of signal reduced in wet weather • quality of signal reduced when not in line of sight • must be in range of base station • must be above ground/close to surface • large buildings/installations reflect or diffract signals • transmission rate slows with a large number of users. • not a secure system/can be hacked Ignore • health and safety • distance is far/ long • satellite communication

Mark scheme (award up to 6 marks) refer to the guidance on the cover of this document for how to apply levels-based mark schemes*.		
Level	Mark	Descriptor
Level 0	0	No rewardable material.
Level 1	1–2	• Demonstrates adequate knowledge and understanding of scientific facts/concepts to the given context with generalised comments made. • Generic statements may be presented rather than linkages to the context being made so that lines of reasoning are unsupported or partially supported. • The comparison will contain some similarities and differences showing some structure and coherence.
Level 2	3–4	• Demonstrates good knowledge and understanding by selecting and applying some relevant scientific facts/concepts to provide the comparison being presented. • Lines of argument mostly supported through the application of relevant evidence drawn from the context. • Demonstrate an awareness of both similarities and differences leading to a comparison which has a structure which is mostly clear, coherent and logical.
Level 3	5–6	• Demonstrates comprehensive knowledge and understanding by selecting and applying relevant knowledge of scientific facts/concepts to provide the comparison being presented. • Line(s) of argument consistently supported throughout by sustained application of relevant evidence drawn from the context. • The comparison shows a logical chain of reasoning which is supported throughout by sustained application of relevant evidence.
		Total 6 marks



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